

FLOWERS FOR ALGORITHMS: IS THERE AN AI TECH BUBBLE?

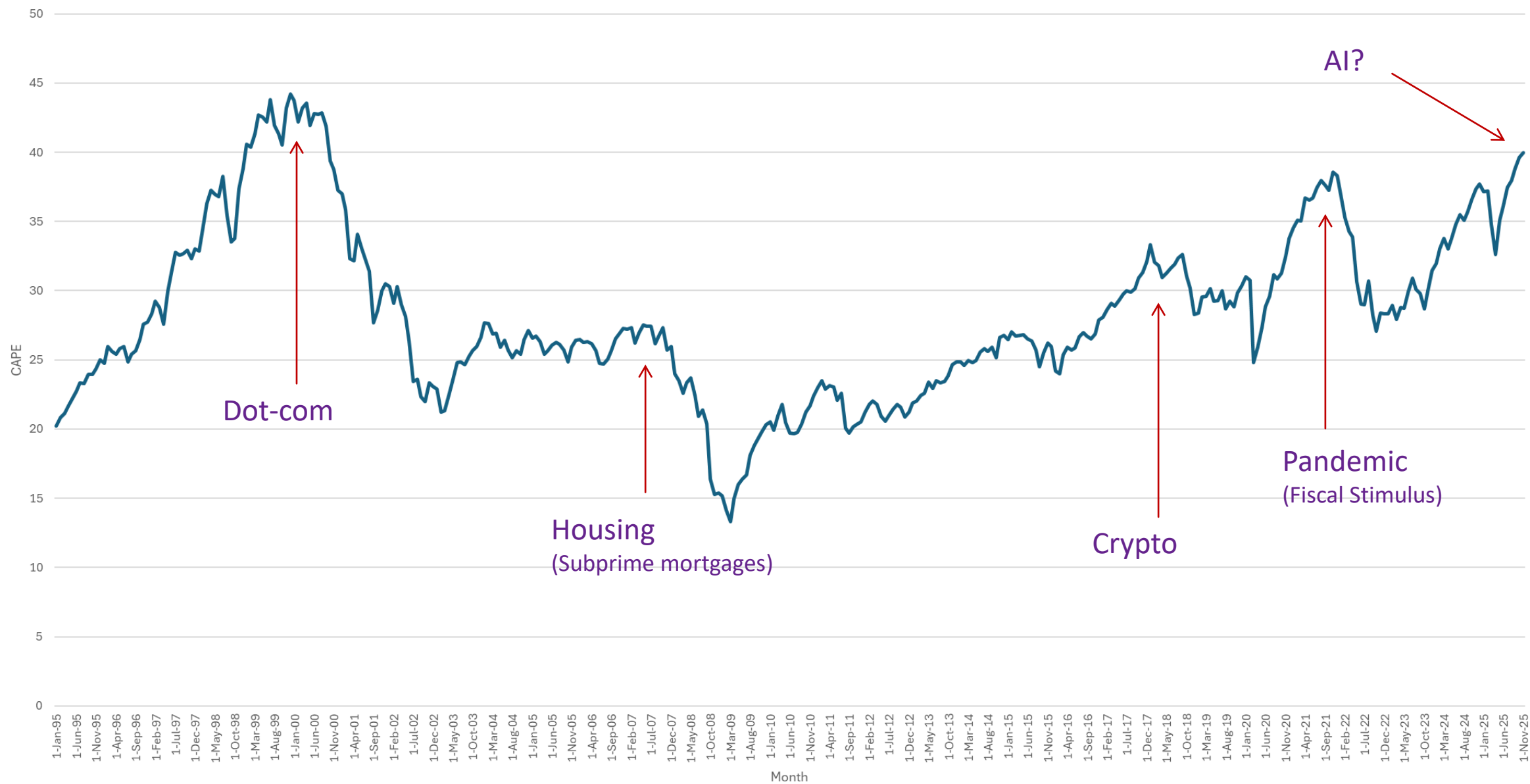


Walter G. Park, American University, December 2025

Outline

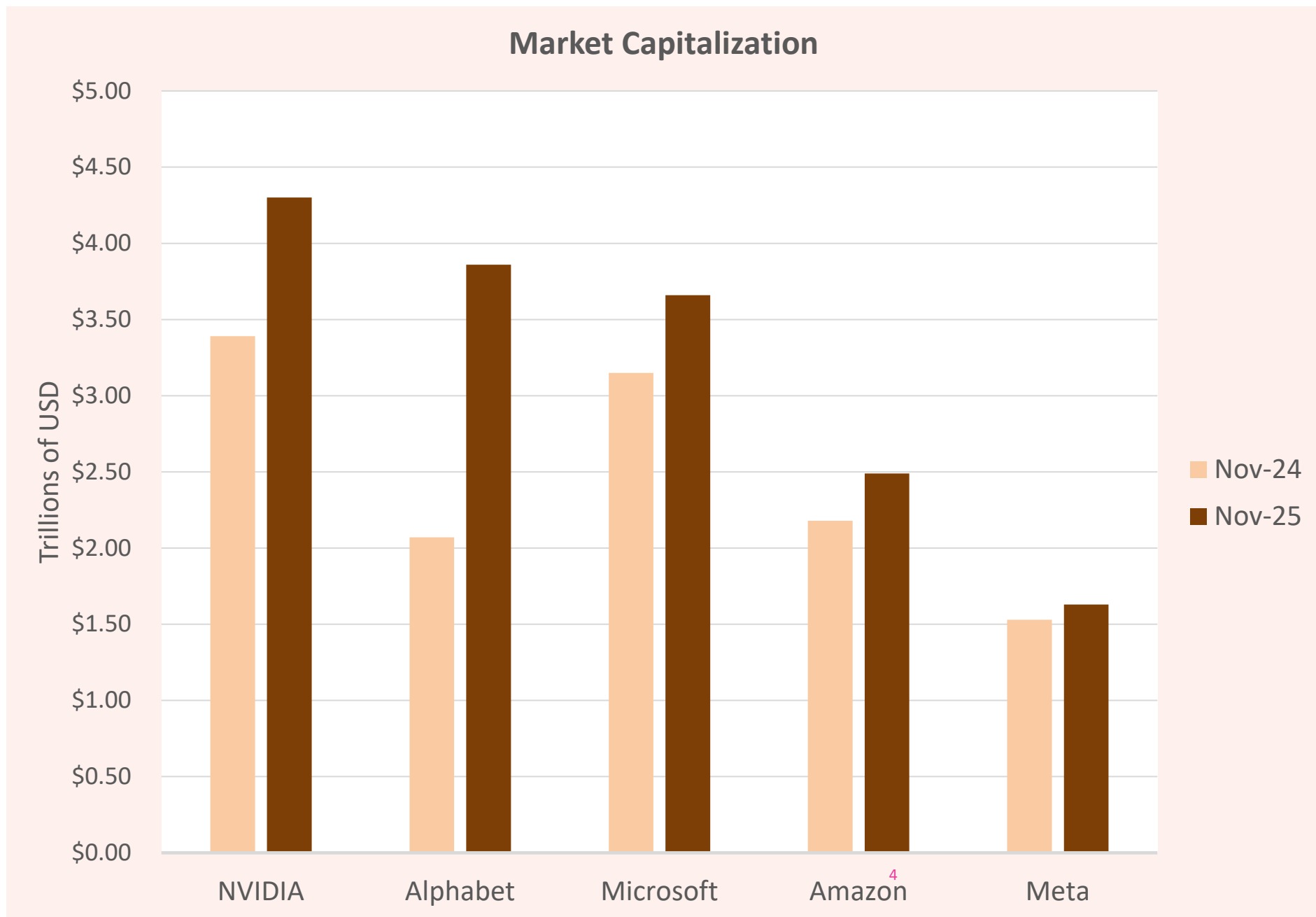
- ▶ Trends in Capital Expenditures
- ▶ Explanations
- ▶ Market Fragilities
- ▶ Market Valuation
- ▶ Technological Fundamentals
- ▶ Competition from Open Models
- ▶ Competition from China
- ▶ Overall Assessment

Cyclically-adjusted Price Earnings Ratio of S&P 500: 1995 - 2025



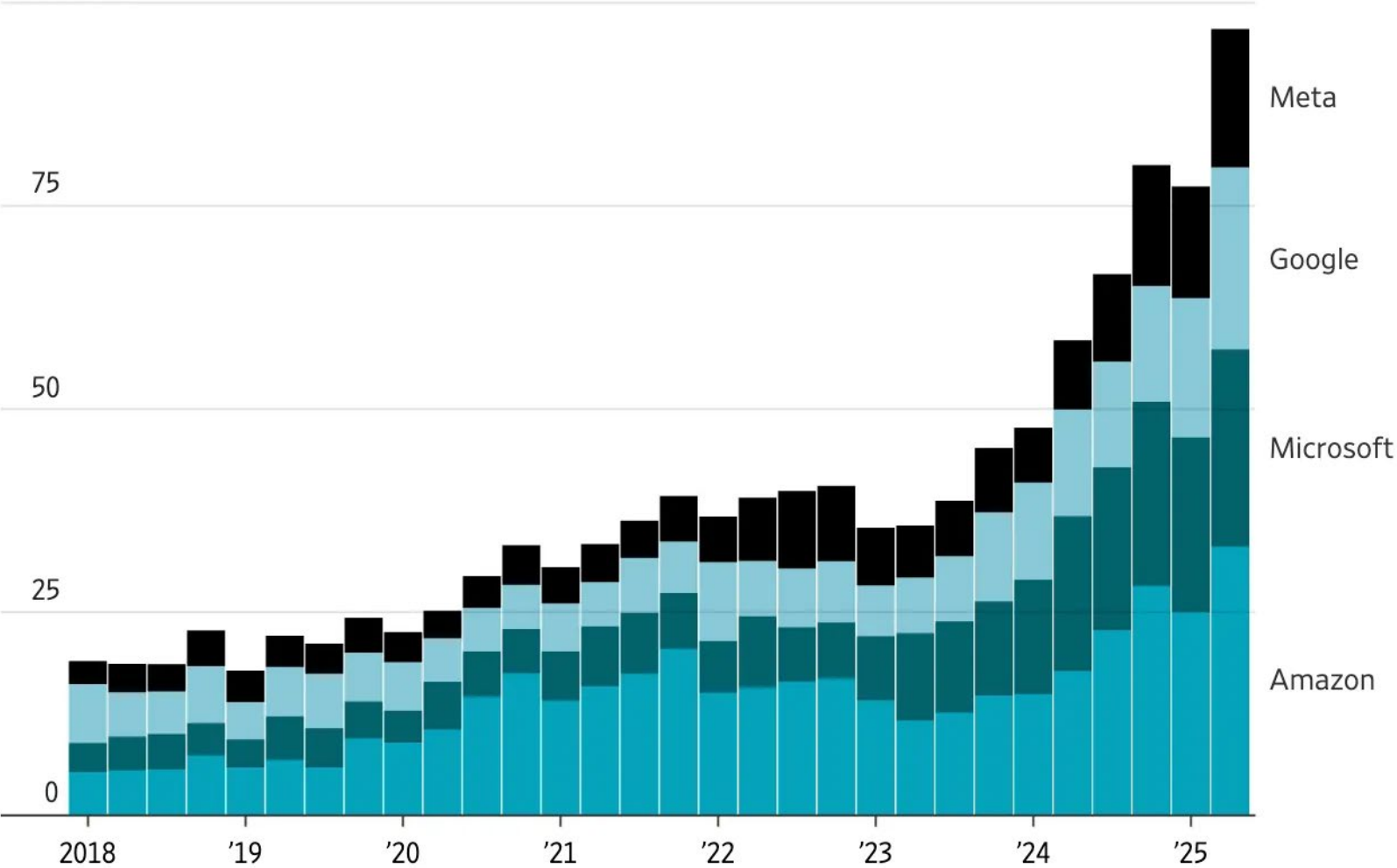
Market valuation

Source: Capital IQ



Capital expenditures, quarterly

\$100 billion



Source:
Forbes.com, Aug 14, 2025

Note: Data are for calendar quarters and include finance leases.
Source: the companies

Capex load

CAPEX/REVENUE RATIO			R&D/SALES RATIO	
Company	2024	2025	2024	2025
MSFT	0.18	0.23	12.0%	11.5%
META	0.24	0.38	26.7%	27.7%
GOOG	0.15	0.24	14.1%	14.4%
AMZN	0.12	0.18	13.9%	14.9%
NVDA	0.02	0.02	14.2%	9.9%

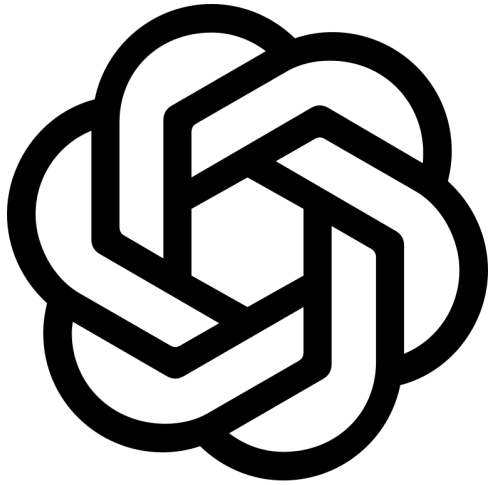
Source: Annual Reports, SEC Filings

INCREASED LEVERAGE

Firm	DEBT Dec 2024	DEBT Sept 2025	%Growth in DEBT	DEBT/ASSET Dec 2024	DEBT/ASSET Sept 2025
META	\$93.4 bn	\$109.8 bn	16.2	0.34	0.36
GOOG	\$124.2 bn	\$149.6 bn	18.6	0.28	0.28
AMZN	\$338.9 bn	\$358.3 bn	5.6	0.54	0.49
MSFT	\$231.2 bn	\$273.3 bn	16.7	0.43	0.43
NVDA	\$32.3 bn	\$42.2 bn	26.7	0.29	0.26

... but Debt-to-Asset ratio is fairly stable

Source: Annual/Quarterly Reports, SEC Filings



RESPONSIBLE OpenAI?

Over 800 Million Active Users of ChatGPT

Only a fraction of them are **paid subscribers**. OpenAI also derives earnings from enterprise clients (workflows, GPT features embedded in products) and API-based licensing.

Estimated Net Loss of \$5 billion (2024)

OpenAI is a private company. Therefore, market values and operating data are not available. Information based on self-reports and news events.

Revenue in 2024 = approx. \$3.7 billion

CEO of OpenAI (Sam Altman) claims revenues are growing and could be \$6.7 bn in 2025.

Yet, CAPEX = \$9.4 bn!

Open AI announced that it will spend \$1.4 trillion more on capital expenditures during 2025 – 2035.

Motivation: Why Are firms Investing so much?

Explanation 1: DERIVED DEMAND

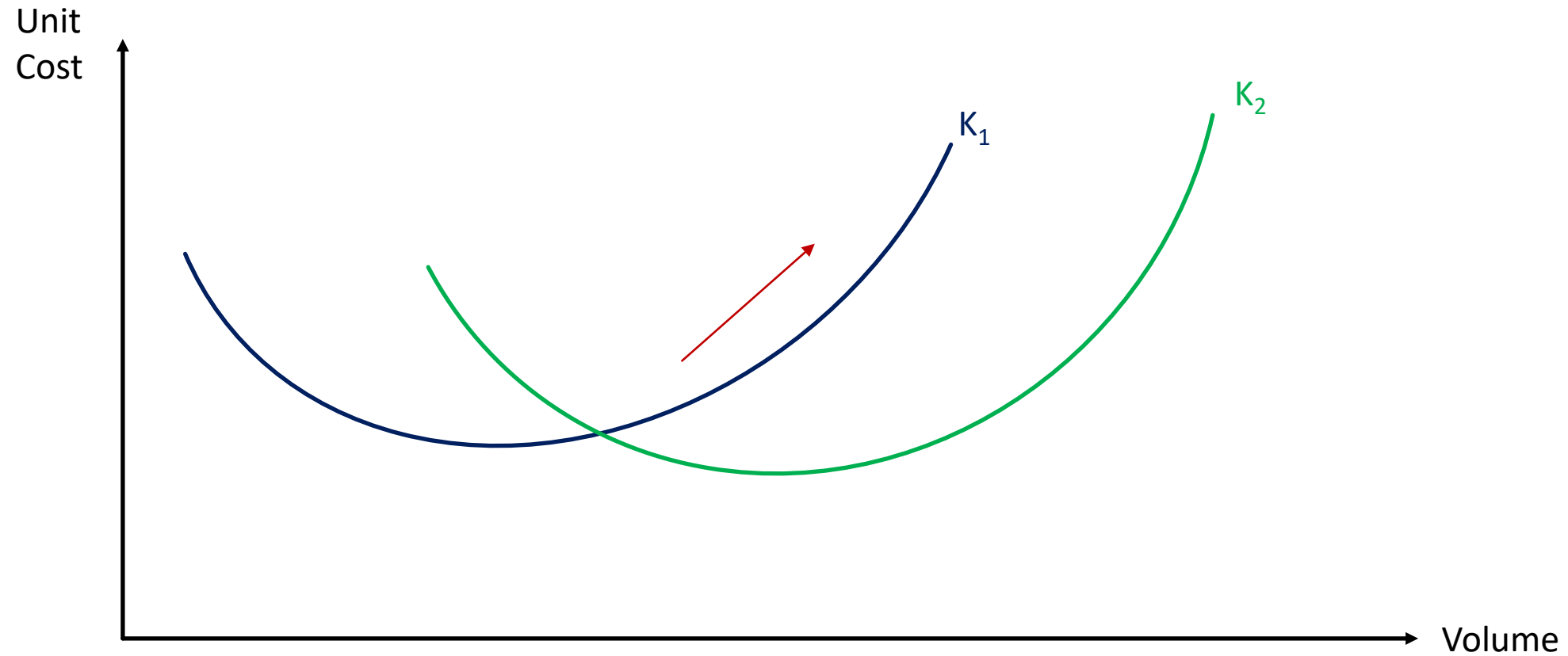
Data-center investment is a consequence, not a cause, of AI adoption. More and upgraded data centers are needed to meet the final demand (of consumers, businesses, and government agencies) for AI services (see next slide).

Explanation 2: JOB CREATION

Employment effects with the building and building of data centers; once built, they are warehouses with mostly maintenance staff.

Capacity expansion (Data Centers)

Meet demand for computing power



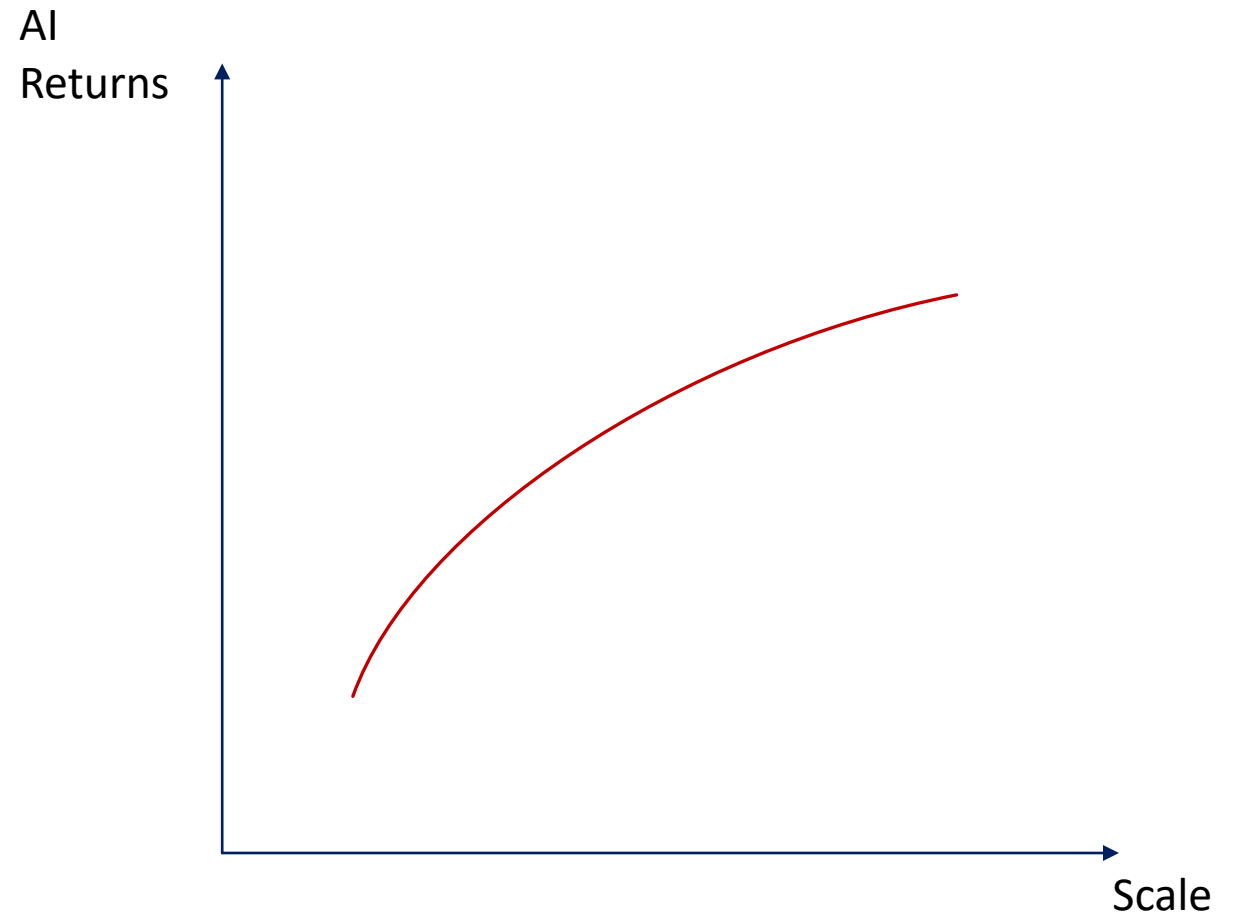
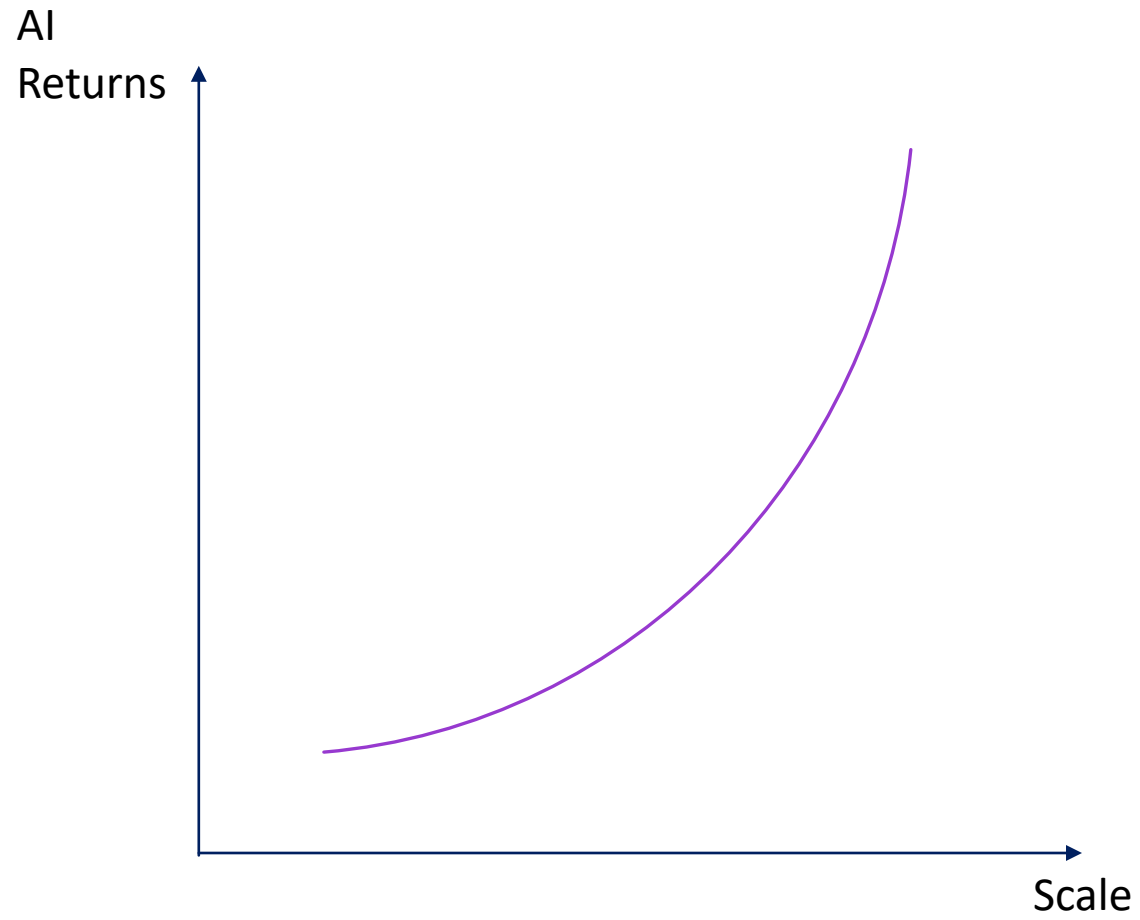
Quest for AGI



Explanation 3: INNOVATION-DRIVEN

- ▶ Companies are under pressure to develop more **foundational** applications of AI: health, medicines, advanced robotics, and scientific discovery.
- ▶ AI companies spend in pursuit of *Artificial General Intelligence* (“With enough breakthroughs, ..., ‘we will actually reach AGI.’” See Karen Hao (2025, p. 176), referring to an OpenAI *Research Road Map*, 2021.)
- ▶ Progression from AGI to superintelligence to a technological singularity.

Returns to scale?



CAN PRIVATE FIRMS* TRIGGER A MARKET CORRECTION?

- OpenAI, Anthropic, xAI, Stability AI, etc.

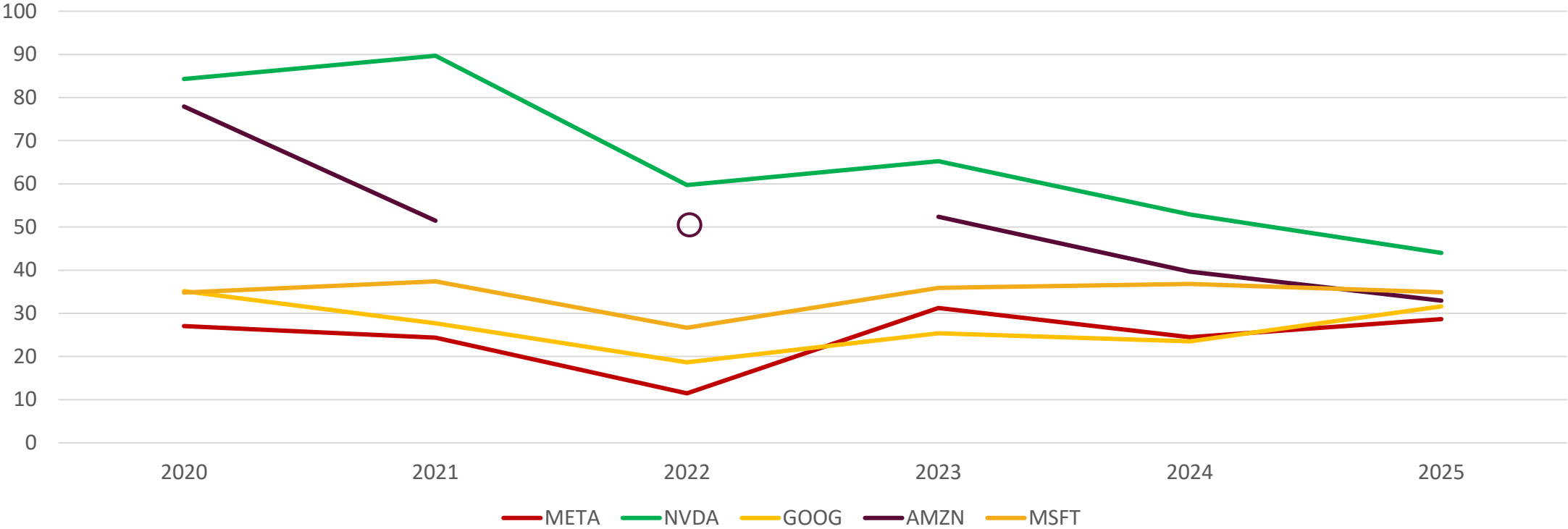
LARGE FIRMS ARE PUBLICLY TRADED

These firms currently post strong earnings, conduct R&D, and patent intensively. Their debt-to-asset ratios < 1 , and their market valuations and current PE ratios are not outliers from the perspective of the past 5 years. See next chart.

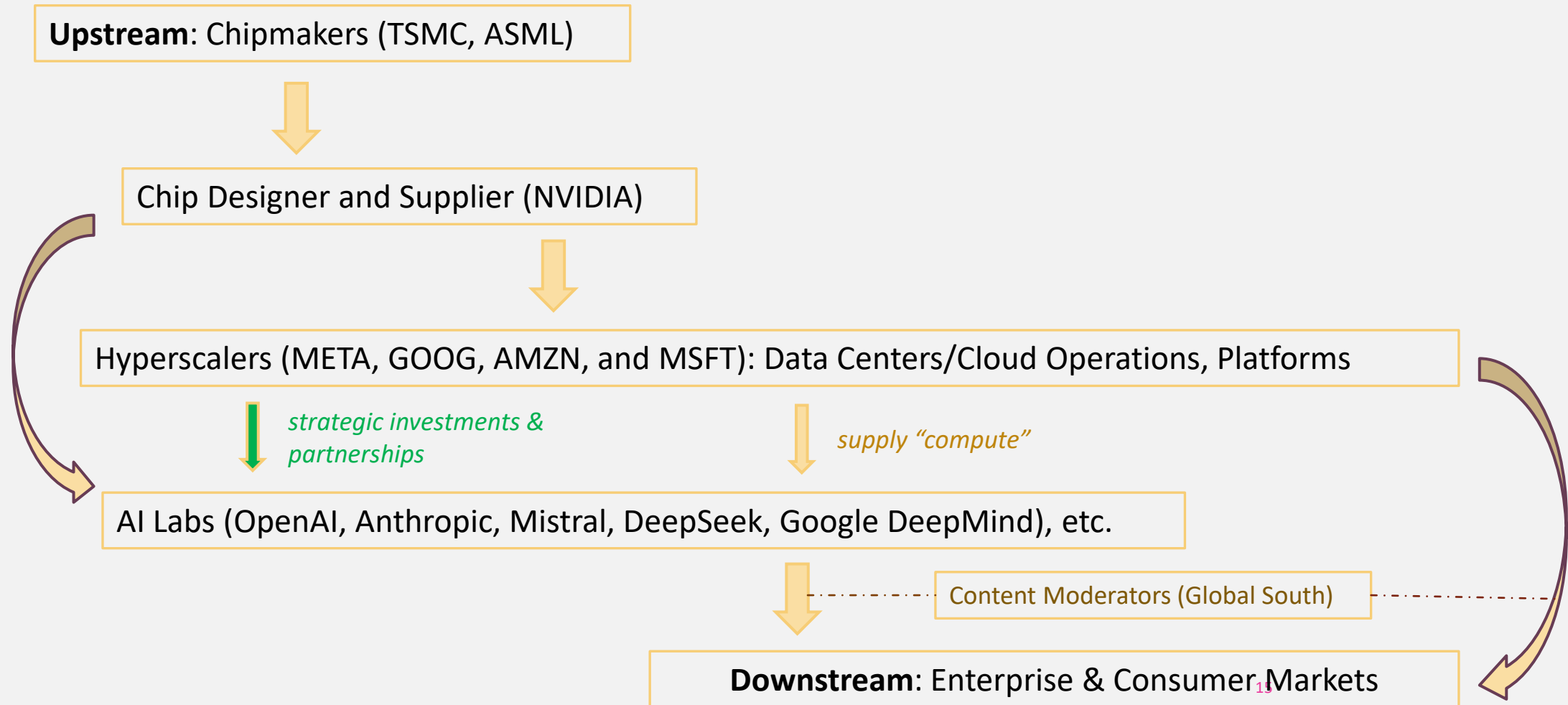
PRIVATE FIRMS ARE NOT ...

Companies, such as OpenAI, have been posting losses, despite investing heavily in compute, being obligated to buying compute from AWS, Azure, GCP, and renting billions in GPUs. Venture capitalists and other funders appear still to place high valuations on these companies. But since these companies are not represented in our stock portfolios or mutual funds, they will not directly trigger market correction; however, they can do so indirectly since they are linked to the publicly traded companies through partnerships, supply chains, and platform technologies.

P.E. Ratios by company (end-of-year)



Supply chains connect private and publicly-traded companies



Market Fragilities

- Suppose OpenAI and other over-leveraged firms unable to pay off debt.
- Negative REVENUE shock to Hyperscalers (GPU warehouses go underutilized)
- Hyperscalers lose revenue stream from compute (AI labs have long term compute commitments to them).
- NVIDIA sees decline in orders.
- Downstream Users dependent on AI experience disruptions in workflow.
- Venture-capital pulls back. Start-ups close or consolidate.
- Supply chain pinpoints how poor financial performance of private AI companies can reverberate to the rest of the AI industry.

Aligning market valuation and fundamentals

How to test for a “bubble”:

1) Descriptive/Narrative Approach

Comparative analysis. L. Floridi (2024): “... there is an argument to be made that the current AI Hype Cycle shares significant similarities with previous tech bubbles ...”

Technological doubt. S. Storm (2025) argues that we have reached “peak GenAI” and diminished returns, casting doubts on AGI. Floridi also raises the possibility that AI is a new form of alchemy.

2) Rigorous Empirical Analysis

Relate market valuation (V) to fundamentals (F). Overvaluation if $V > F$. *A canonical approach:*

$$V_{it} = f(\tilde{F}_{it}) + \varepsilon_{it}$$

Analyze behavior of residual, ε_{it} , where i indexes firm and t time. Check for divergence. Non-stationarity.

Fundamentals and Technological Innovation

Number of AI Patents Granted

Year	Total (000's)	USA	CHINA	EUROPE	INDIA	ROW
2012	5.2	40.0%	13.4%	0.1%	8.6%	37.9%
2013	6.3	32.4%	18.8%	0.1%	7.9%	40.7%
2014	6.8	34.6%	21.2%	0.2%	7.8%	36.2%
2015	7.8	38.1%	20.4%	0.2%	5.9%	35.4%
2016	10.2	42.1%	18.1%	0.4%	6.3%	33.1%
2017	13.4	42.8%	25.8%	0.4%	6.4%	24.6%
2018	17.3	39.2%	31.3%	0.5%	6.2%	22.8%
2019	27.0	34.5%	36.9%	0.4%	6.6%	21.6%
2020	43.4	28.7%	43.8%	0.5%	6.1%	20.9%
2021	66.1	31.2%	42.9%	0.6%	5.3%	20.0%
2022	94.6	26.5%	50.2%	0.5%	4.1%	18.7%
2023	122.5	21.0%	57.9%	0.5%	3.5%	17.1%

Source: HAI Stanford, 2025 AI Index

% World Share

Number of AI Publications in Computer Science

Year	Total (000's)	% World share					% Organization share				
		USA	India	Europe	China	ROW	Academia	Industry	Gov't.	Nonprofit	Other
2013	101.9	11.8%	4.8%	20.3%	17.2%	20.4%	86.0%	6.6%	4.5%	1.9%	1.0%
2014	104.4	11.3%	6.0%	20.3%	16.2%	20.7%	85.8%	6.3%	4.9%	2.1%	0.9%
2015	105.7	11.6%	6.7%	20.1%	14.9%	21.7%	85.4%	6.6%	5.2%	1.9%	0.9%
2016	107.3	11.7%	7.3%	19.8%	14.0%	21.9%	85.6%	6.6%	4.9%	2.0%	0.9%
2017	116.9	12.0%	7.1%	19.0%	14.8%	22.4%	85.3%	6.9%	4.8%	2.0%	1.0%
2018	139.7	12.3%	6.7%	18.1%	16.0%	22.2%	85.0%	7.4%	4.6%	2.0%	1.0%
2019	164.2	12.8%	6.0%	18.1%	17.6%	22.7%	83.9%	8.5%	4.6%	1.9%	1.1%
2020	181.1	12.9%	6.6%	17.8%	17.9%	23.1%	83.6%	8.6%	4.5%	2.1%	1.2%
2021	204.1	12.4%	7.4%	17.7%	19.7%	23.3%	83.5%	8.8%	4.6%	1.9%	1.2%
2022	202.8	9.8%	9.0%	16.0%	23.5%	23.6%	84.9%	7.2%	4.8%	1.8%	1.3%
2023	242.7	9.2%	9.2%	15.2%	23.2%	22.5%	84.9%	7.1%	4.9%	1.7%	1.4%

Source: HAI Stanford, 2025 AI Index

Fundamentals and Open Innovation

Training and openness

Number of notable AI models by Training code access type

Year	Open (noncommercial)	Open (restricted use)	Open source	Unknown	Unreleased
2014	1		1	30	
2015			2	21	5
2016	2		2	37	9
2017	1		6	40	11
2018	1		16	19	15
2019	11	1	33	1	26
2020	2		22	6	24
2021	3		29	14	29
2022	4		16	38	28
2023	5	2	13	48	37
2024	2	2	2	18	37

Source: HAI Stanford, 2025 AI Index

COMPETITION FROM OPEN MODELS

- ▶ Interesting working paper by Nagle and Yue (2025)
- ▶ Unique dataset from **Open Router**, platform for LLM inference services.
 - ▶ Open Router provides a single API (application programming interface) for developers to access multiple LLMS from many providers (of closed, proprietary and open models).
 - ▶ It charges a small fee on usage credits that developers buy. They spend the credits on inference activity across the many providers. Open Router does not add a markup on what providers charge.
- ▶ This platform can provide data on usage and pricing (per million tokens) by type of model (closed or open). The authors then match the data on models to data on metrics of their performance.
- ▶ Three key findings (overview):
 - ▶ Open models are cheaper (one-sixth the price of closed models)
 - ▶ Open models perform almost as well as closed models (around 90%)
 - ▶ Yet users predominantly use closed models, not open (about 80% of token usage is in closed models)
- ▶ Implication: Consumers are missing out on unrealized savings by not switching
- ▶ Puzzle: Why underutilization occurs? *Paper doesn't test for explanations.*
 - ▶ Possibilities: branding, switching cost (lock-ins), trust and reliability, open models are for experts.

Examples

► Closed Models

OpenAI

Anthropic

Google

xAI

► Open Models

Llama (Meta)

DeepSeek

Mistral

Granite (IBM)

Falcon (Apache)

Qwen (Alibaba)

Closed models (e.g., GPT-4, Gemini, Claude) are accessible only via proprietary APIs, whereas open models (e.g., DeepSeek, Mistral, Qwen) release weights that allow competitive third-party inference.

Ecosystem

- ▶ Model providers and/or inference providers
- ▶ Model development (LLMs) begins with architectural design and training – curating massive datasets (containing trillions of tokens) and running on thousands of high-performance processors (GPUs).
- ▶ Outcome: A model with billions of numerical parameters (called “weights”). A weight is a numerical parameter inside a neural network that determines how strongly one unit influences another unit. During training, weights are adjusted so the model learns patterns (language, reasoning, math, and so on). Together, all weights encode the model’s “knowledge” and behavior.
- ▶ Business decision is next: keep the model proprietary (weights, architecture, training data are trade secrets) OR release as an open model.
- ▶ First-party providers (**who train and serve their own models**) versus Third-party Providers (**who serve models trained by others**). Each provider maintains its own APIs.
- ▶ What is the value-added of third-parties? They focus not on exploration but on optimization. Enterprises may take base model and use open weights to fine tune it to perform specialized functions.
- ▶ Note free ≠ no cost. Open models still need chips, servers, electricity, space. Cf. Free car vs. Taxis.
- ▶ Third-party developers can create applications for enterprise CRMs and specialized tasks using the **API** of AI models. Applications can be sold to downstream markets. These applications add market value beyond model inferences.

Testing Value and Performance

- ▶ Application Programming Interface (API) = doorway to a model (allow for batch processing, automated tasks, simulations and experiments, and building of agents). APIs don't allow editing of a model's knowledge base or change outputs.
- ▶ Let Price = Number of Input Tokens $\times p_{\text{input}}$ + Number of Output Tokens $\times p_{\text{output}}$
where $p_{\text{input}} < p_{\text{output}}$ since generating output is computationally heavier than reading input
- ▶ The above can be interpreted as the price of one inference or one API request. Tokens are used in a prompt, and they are used when AI completes its tasks (i.e., generates a response).
- ▶ Performance Metrics (Nagle & Yue)
 - ▶ Artificial Analysis – a platform that measures and compares performance of different AI models. Evaluates ability to answer PhD level science questions (e.g., a GPAQ score) and ability to generate code.
 - ▶ LM Arena – human-based scores (i.e., users vote on which responses they prefer)
 - ▶ Criticisms: the performance metrics are not thorough. They don't capture actual errors (e.g., RMSPE). They don't measure degree of "hallucinations", etc.

FINDINGS

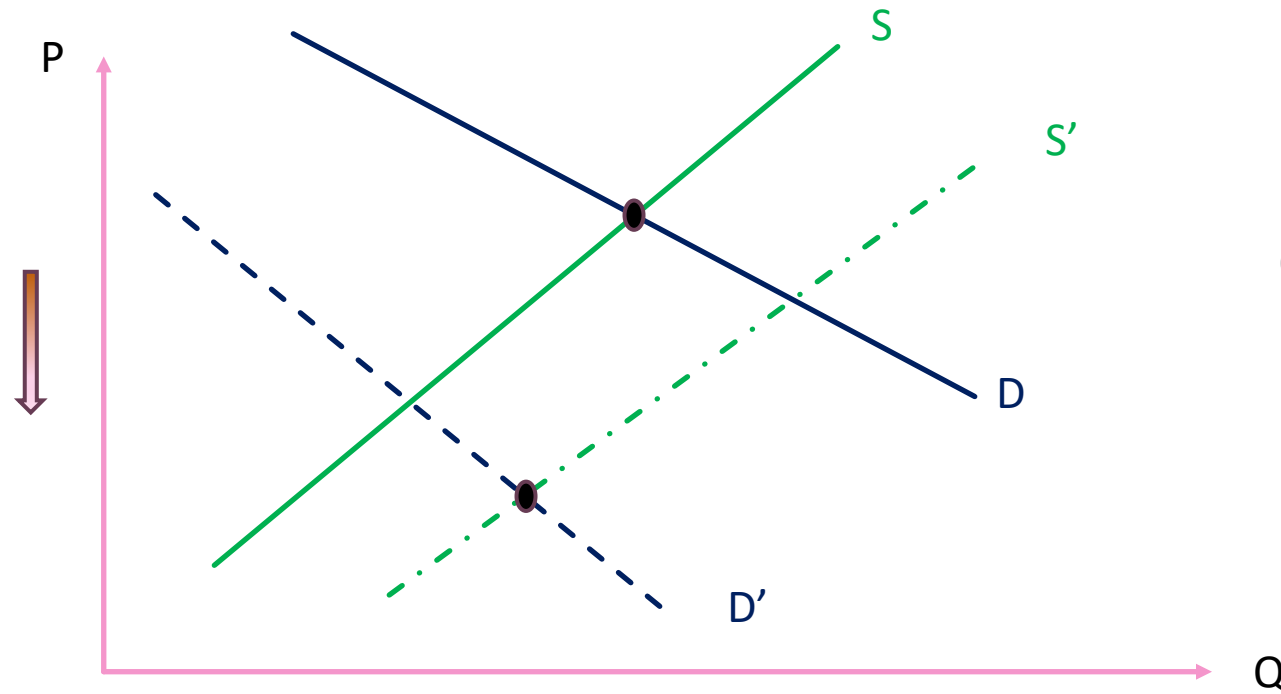
- ▶ Open models have on average: 20% of token usage
- ▶ They have 5% share of revenue
- ▶ Their price is 10 – 30% lower (and declines over time upon release)
- ▶ Their performance is about 90% of closed models
- ▶ Fast catching-up (of open models to closed models' performance)
 - ▶ During Jan – June of 2024, the average catch-up time = 27 weeks
 - ▶ July – Dec 2024, avg catch-up time = 17 weeks
 - ▶ Jan – June 2025, avg catch-up time = 13 weeks
- ▶ Sources of catch-up: knowledge spillovers and rapid imitation. Open models specialize in engineering efficiency.
- ▶ Consumers could save \$25 bn annually by switching from closed to open models

But user demand for OPEN MODELS is lower, controlling for price and performance differences:

$$\ln Token_m = \beta_1 \ln P_m + \beta_2 \ln PERF_m + \beta_3 \ln OPEN_m + \varepsilon_m$$

Study finds that $\hat{\beta}_3 < 0$. Point estimate suggests usage of Open Models is 60-80% lower.

Note: Analysis is not structural ...



Open Model Market

- Price Gap: $\Delta = P^{\text{Closed}} - P^{\text{Open}} > 0$
- What is the compensating differential dependent on?

Explanations for $\Delta > 0$

- ▶ Study mismeasured performance of Open Models
- ▶ Consumer Preference
 - ▶ Branding/Reputation of Established Proprietary companies
 - ▶ Security risks (perceived or real) associated with open models (data exfiltration, compliance with safety)
- ▶ Switching Costs (or lock-in)
- ▶ Misconceptions about Open and/or Foreign AI Models
- ▶ Future hypotheses: Open models dominate under low liability workloads, when service quality gaps are smaller. US government tends to use proprietary models.

Implications

- ▶ At least open models may **discipline** the pricing of closed, proprietary models (keep prices in check).
- ▶ Open models have relevance for the **market valuation** of closed, proprietary models. Competition tends to dissipate the profits of the latter.

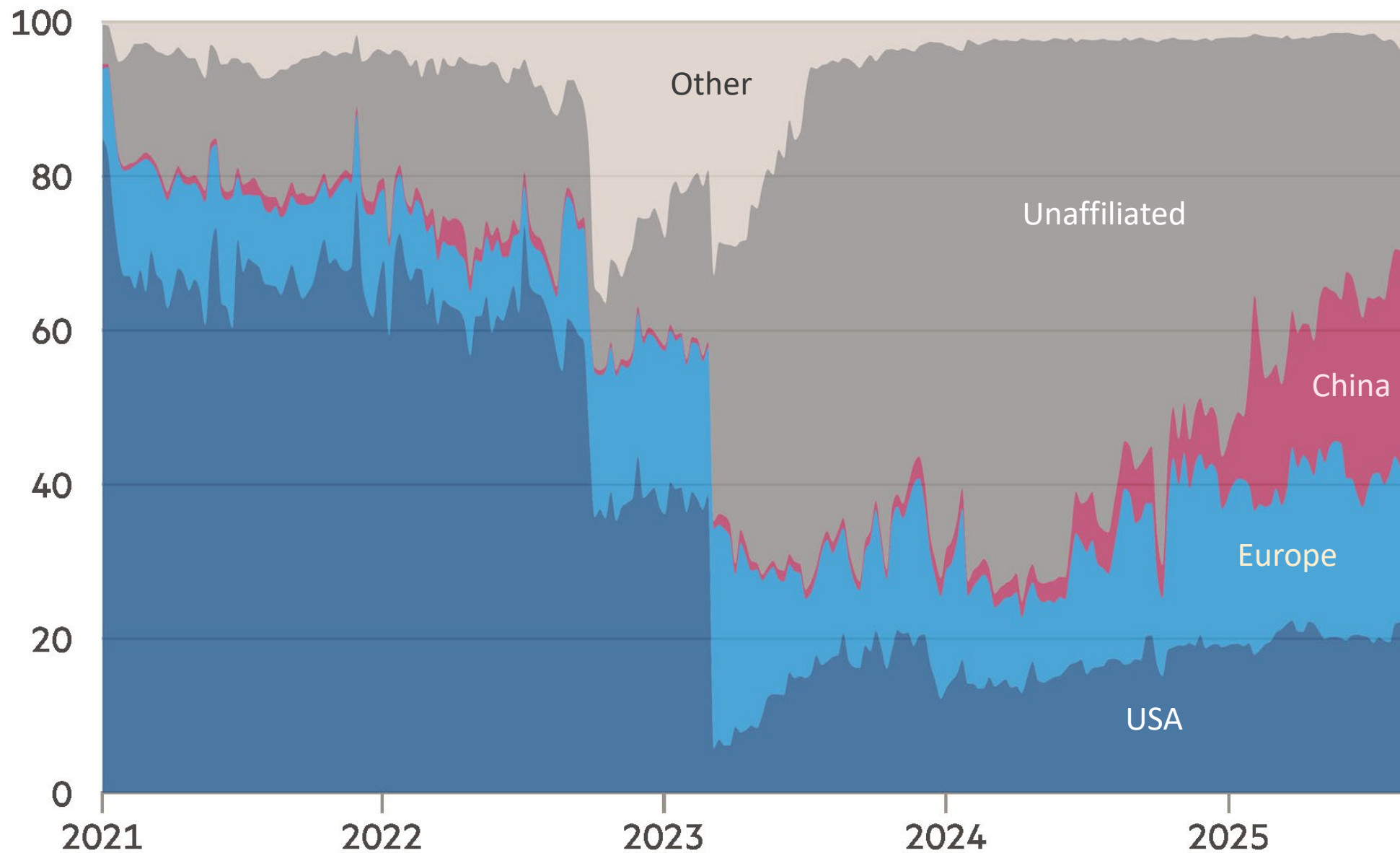
$$V_0 = \pi_0 + \frac{\pi_1}{(1+r)} + \frac{\pi_2}{(1+r)^2} + \frac{\pi_3}{(1+r)^3} + \dots + \frac{\pi_n}{(1+r)^n} + \dots$$

- ▶ A robust Δ should allow closed models to recoup their early investments ...
- ▶ Why release open weights?
 - ▶ Open models help seed adoption and diffusion, create standards
 - ▶ Open communities attract top talent. Open models generate learning-by-doing. Encourage innovation that can reduce development and operation costs and improve capabilities.
 - ▶ Proprietary firms earn revenue from complementary assets (compute, hardware, chips, cloud hosting, tech support, and other adjacent services)

Competition from China

- ▶ China overtakes USA in the global market for “open” artificial intelligence models in terms of downloads of new models. DeepSeek and Alibaba models are the most popular downloads of Chinese models.
- ▶ Chinese companies have some cost advantages in computation (lower electricity, state subsidies for compute, models with fewer parameters)
- ▶ Strategic trade policy: Release open models to drive away market share of big US tech companies, which rely upon on proprietary control for profitability.
- ▶ Quicker release strategies: Chinese companies release weekly or bi-weekly, rather than every 6 months or a year like US labs. Their strategy is to distribute widely (access), while US companies’ strategy is focused on bigger aims, like state-of-the-art or AGI.
- ▶ Thus, geopolitical factors are:
 - ▶ Reduce dependence on proprietary APIs, especially those of the U.S. Disclose weights to avoid firms from monopolizing a technological innovation.
 - ▶ Open releases accelerate diffusion of Chinese capabilities and designs
 - ▶ React to export controls of the U.S. Find ways to compete better globally.

Weekly national download share (%)



Source: [Melissa Heikkilä](#), Financial Times, Nov. 25, 2025. Data from Hugging Face

Discussion Questions

1. Is there an AI hype? Has the technology fallen short of expectations?
2. Will the demand for closed (proprietary) AI remain strong?
3. Are open models “underutilized,” or are users rationally paying a premium for something we don’t observe?
4. Why do some firms invest heavily in training large AI models and then release them as open models, rather than keeping them proprietary and monetizing inference directly?
5. Will proprietary AI firms vertically integrate with open-source communities?
6. Is China’s strategy of releasing open models best understood as competition, trade policy, or something else?